

Brain Tumor Classification: Feature Fusion

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Abstract—Brain tumor detection is a challenging task in medical image analysis. The manual process performs through domain specialists is a more time-consuming task. Numerous works are represented for brain tumor detection and discrimination but still, there is a need for a fast and efficient technique. In this article, a technique is presented to distinguish between benign and malignant tumor. Our method integrates the enhancement of the lesion, features extraction, selection, and classification methods. In-lesion enhancement phase, the N4ITK3 method is applied to normalize the input image. A serial fusion based technique is used for feature extraction and fusion by using HOG (shape), SFTA and LBP (texture) based features. The fused features are selected through the Boltzmann entropy technique. The fused feature vector is supplied to the multiple classifiers to compare the better prediction rate. The three publically available BRATS databases are utilized for tumor detection. The method is trained separately on benchmark databases. The performance outcomes demonstrate that the mean dice similarity coefficient (DSC) is 0.99 for tumor estimation. The performance outcome illustrates that the proposed model effectively classifies the abnormal and normal brain region. The comparison performed with previous techniques proves the effectiveness of the presented technique.

Keyword. Glioma, MRI, Entropy, Texture, Shape

I. INTRODUCTION

Brain tumor appears due to the growth of abnormal cells [1, 2]. Gliomas are the main type of tumor in adults. Gliomas grow from glial cells and might be discriminated into the grade I to IV. On this scale, a benign tumor belongs to grade I and grade II [4][5], malignant tumor belongs to grade III and grade IV [5][6]. Detecting the brain tumor at an initial stage significantly increases the possibility of the tumor patients survival rate [7]. Magnetic resonance imaging (MRI) is used to evaluate soft tissues in the human body[3]. Moreover, multimodal MRI is used to analyze the tumor blood-brain barrier (BBB) integrity, vascularity, and cellularity. The different sequences of the MRI protocol give crucial opposite information. In T1 contrast enhancement images, borders of the tumor region appear brighter due to the contrast accumulation in the vessels of the brain region. In these images, active and necrotic tumor regions might be detected more easily. In T2, the edema region appears brighter. Flair is used to separating the edema region from the CSF [4, 5].

A. Motivation

Brain tumor detection is a more difficult task due to larger variations occurs because of the texture and shape of the tumor region and discontinuity, irregularity and unclear gliomas boarder. Manually analyzing the medical images is an error-prone and time-consuming process. Feature extraction and selection is also a more challenging task which does not provide a guarantee to achieve always good results [6-8]. The high dimension of extracted features is also a serious problem for brain tumor detection. A number of features increase the correlation between features and therefore, in turn, increases complexity between the detection of tumor labels. This problem seriously degrades the classifier's accuracy and affects computational time. The selection of an appropriate number of testing and training images is hard to calculate the performance of predictors. In this manuscript, we propose an automated feature extraction and selection method to classify the tumor region more accurately.

B. Major Contribution

The major contributing steps in this manuscript are defined below:

1) The proposed method consists of the training phase in which classifier learns characteristics of the input images on the basis of the shape of the tumor, the classifier recognizes the type of the tumor in a test image. The classification is performed by combining the multi MR sequences instead of using a single MR modality independently.

2) The proposed method consists of four steps: lesion enhancement, feature (extraction/selection/fusion) and classification (performed on the basis of multiple classifiers). 10-fold cross-validation is used to test presented technique accuracy.

3) The efficient and accurate features extraction and the selection is vital that affects directly the classification outcomes. The texture (SFTA, LBP) and shape based (HOG) features are utilized because the tumor appears in a different shape and appearance. Therefore, these extracted features provide better discrimination results between malignant and benign tumor.

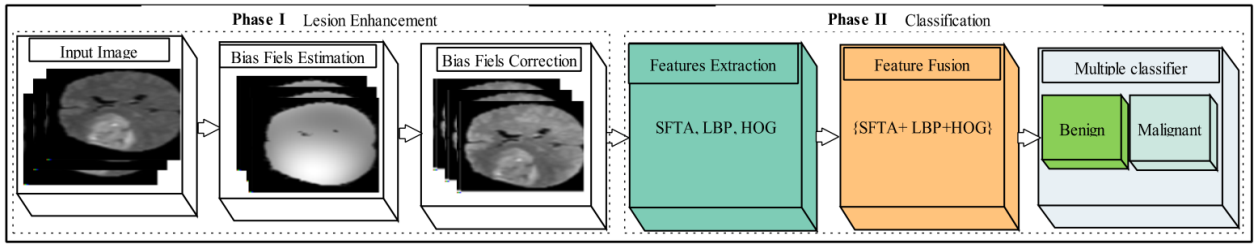


Fig. 1. Proposed Research Layout

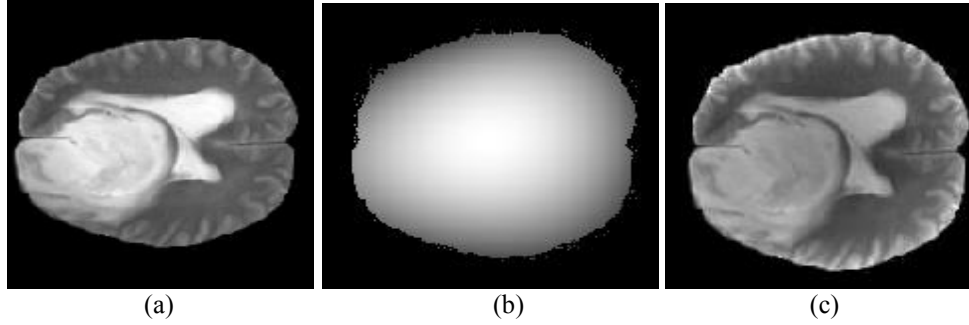


Fig. 2. Image Normalization (A) Input Image (B) Bias Field Estimation (C) Bias Field Correction

The manuscript organization is as follows: related work discusses in section II. Proposed method steps are elaborated in section III and evaluation results are presented in section IV. The manuscript conclusion is described in section V.

I. RELATED WORK

Many research works are performed for the detection of a brain tumor in the last ten years. This research work discusses methods that are used for tumor detection. Brain tumor detection methods include generative and discriminative methods for differentiating between the brain images [9-11]. Generative methods are dependent on the domain-specific related to normal and abnormal brain tissues. Identification of a tumor region is very challenging work because tumors appear in different portions of the brain region with different sizes and shapes [12]. Commonly, these methods are dependent on the anatomical methods that are acquired after aligning 3D atlas image or computed template from many normal brains. The generative method is used for computing the posterior probability of healthy tissues [13]. Active contour model is also utilized to determine the probability of the abnormal region [14]. Several active-contour techniques with brain symmetry and features alignment are used for analysis of the brain abnormality [15-17]. Image registration and segmentation steps are performed to detect brain tumor [28, 34]. Other techniques are dependent on discriminative methods including low level features [18, 19], Gabor filter banks [20, 21], local histograms [22], SVM[23-25], and decision forests(RF) [26] are utilized for detection of brain tumor [27].

II. METHODOLOGY

In this manuscript, a method is suggested for brain tumor classification. The presented technique framework is mentioned in Figure 1. The proposed technique accurately classifies the brain tumor into their corresponding classes. The proposed technique is applicable to all sequences of the MR images. The proposed technique advantage is to achieve maximum classification results.

A. Preprocessing

The noise appears in the MRI due to the magnetic field fluctuations in the coil. Several issues arise in the classification of the brain images such as bias field estimation due to intensity inhomogeneity. Normalizing the image intensities is a more vital step in the analysis of human brain MRI. Therefore in this manuscript, N4ITK bias field correction method [28] is utilized in the input image based on their intensity variation. It provides better results to normalize the MR images that are mentioned in Figure 2.

B. Proposed extracted feature and selection method

Extraction of features is a more vital phase for image representation. The total 161 features are concatenated into the three categories including 100 (HOG), 59 (LBP), and (21) SFTA. Figure 3 shows the feature extraction and selection method.

C. SFTA Texture Features

In texture image analysis, SFTA features are used due to robustness and low computationally cost. SFTA features are extracted in the two phases (a) fractal dimension by normalizing the boundary of the input image with gray level size and mean is computed (b) the complexity of tumor boundary is defined by fractal measurements. The fractal (D) dimension is measured through a counting box method [29].

D. HOG

HOG (shape) based [30] features are used for discrimination between healthy/unhealthy slices with (1×3780) dimension feature vector. In which [30, 31], divide the image into smaller parts and combined these parts. The segmented images are filtered by Sobel kernel function to achieve gradient with the directions of G_x and G_y . At every pixel calculate angle and magnitude in Equation (1) and Equation (2).

$$f_{HOG|G(i,j)} = \sqrt{G_x(i,j)^2 + G_y(i,j)^2} \quad (1)$$

$$f_{\text{HOG}_{\theta_G(i,j)}} = \tan^{-1}\left(\frac{G_y(i,j)}{G_x(i,j)}\right) \quad (2)$$

Where $|G|$ denotes gradient magnitude, θ_G represents gradient angle, i and j denote columns and rows.

In $|G|$ represent magnitude gradient, θ_G denotes gradient angle, and rows and column are represented by j and i .

Based on a gradient, angle allocates cell votes to bins.

On the basis of the gradient, angle allocates cell votes to the bins. Finally, obtain the normalization vector by using each block of the histogram. HOG feature descriptor is employed on normalized images that are obtained after preprocessing with 08 bin cells and represents in Equation (3) below.

$$f_{\text{HOG}_V N_i} = \frac{V_i}{\sqrt{(\|V\|_2^2 + \epsilon^2)}} \quad (3)$$

where ϵ is a small constant value and might not divided by the zero. V denotes nonnormalized vector in which all histograms in a block are included. Then combined all these

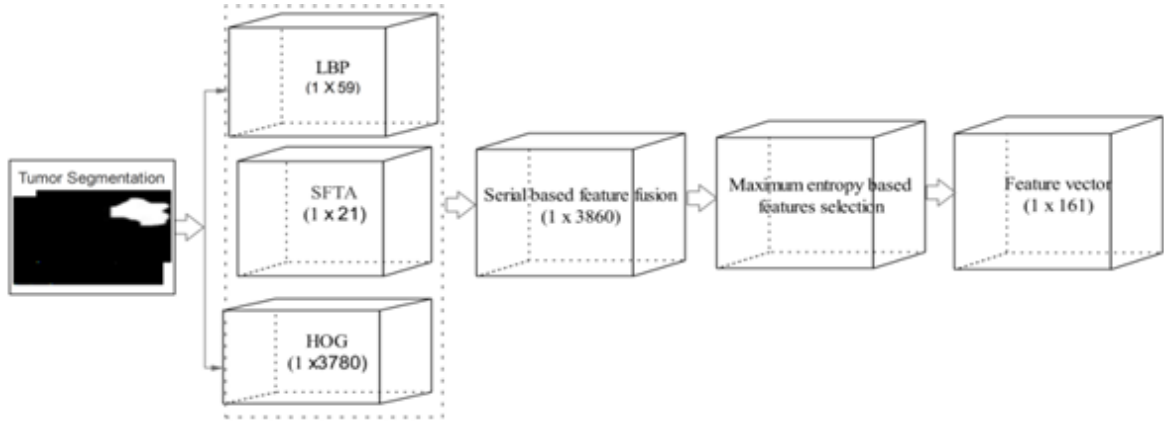


Fig. 3. Proposed Feature Fusion Method

$$f_{\text{LBP}_{1 \times p}} = \{\text{LBP}_{1 \times 1}, \text{LBP}_{1 \times 2}, \text{LBP}_{1 \times 3}, \dots, \text{LBP}_{1 \times n}\} \quad (6)$$

Moreover, these features are concatenated and fused in one vector.

$$\text{Fused}(\text{feature vector})_{1 \times q} = \sum_{i=1}^3 \{f_{\text{HOG}_{1 \times n}}, f_{\text{SFTA}_{1 \times m}}, f_{\text{LBP}_{1 \times p}}\} \quad (7)$$

Where $q \in (n + m + p)$. Fused vector f size is (1×3860) . The entropy-based feature selection [33] method is then applied to each vector; it chooses maximum features [34] on the basis of their score. The explanation of the feature selection method mathematically can be defined in Eq (1-4). We use the score of the entropy. Entropy is used to select a maximum of 161 score-based features from the 3,860 features.

$$B_E = -NE_p \sum_{i=1}^n p(f_i) \quad (8)$$

$$F_{\text{select}} = B_E(\max(f_i, 161)) \quad (9)$$

In Eq (8-9) p denotes the extracted features probability, B_E represent entropy and F_{select} denote selected extracted features, which are 1×161 . Finally fed the selected features into the multiple classifiers for discrimination of the benign and malignant tumor.

G. Evaluation protocol

The presented method is evaluated on challenging datasets including BRATS 2013, BRATS 2014 and BRATS 2015 datasets. BRATS 2013 contains ten LGG, twenty HGG is utilized in training and ten both LGG and HGG are utilized

normalization vectors in each block, the HOG feature vector is achieved.

E. LBP Features

Local binary patterns (LBP) [32] are used for classification purpose. The LBP feature dimension is 1×59 .

F. Feature Fusion

In this article more than features are fused through maximum entropy. The accurate results are achieved by fusing features such as HOG (shape-based) and texture (LBP, SFTA). The three vectors are defined as: f_{HOG} , f_{SFTA} , and f_{LBP} . In which $1 \times n$, $1 \times m$, $1 \times p$ represent the dimension of the f_{HOG} , f_{SFTA} , and f_{LBP} respectively. Then the vectors are then defined as:

$$f_{\text{HOG}_{1 \times n}} = \{\text{HOG}_{1 \times 1}, \text{HOG}_{1 \times 2}, \text{HOG}_{1 \times 3}, \dots, \text{HOG}_{1 \times n}\} \quad (4)$$

$$f_{\text{HOG}_{1 \times m}} = \{\text{SFTA}_{1 \times 1}, \text{SFTA}_{1 \times 2}, \text{SFTA}_{1 \times 3}, \dots, \text{SFTA}_{1 \times n}\} \quad (5)$$

in the testing stage [35]. BRATS 2014 and 2015 contains three hundred and eighty-four images [9, 36]. The proposed technique is validated on nine classification methods includes Decision Trees (DT) [37], Bagged Trees (BT) [38], Linear Discriminant Analysis (LDA) [39], Quadric Discriminant Analysis (QDA) [39], Quadric-SVM (Q-SVM) [40], Cubic-SVM (C-SVM) and linear SVM [40]. Four performance measures include sensitivity, dice similarity coefficient, specificity, and accuracy, are used for analysis of the proposed method.

The experiment is implemented on Matlab 2018a, Core i7 3.4 GHz and 8GB RAM.

H. Experimental Results

The prediction outcomes of multiple classifiers are presented in this section. The results are performed on a 10 fold cross-validation method. The feature fusion process is performed on the entropy-based feature selection method is mentioned in Table I-III.

TABLE I. MAXIMUM ENTROPY-BASED FEATURES SELECTION RESULTS ON BRATS 2013

Classifiers	SE	SP	ACC	DSC
DT	84.6%	91.8%	89.2%	0.85
EBT	90.8%	99.3%	96.0%	0.94
LDA	89.7%	98.6%	95.2%	0.93
QDA	86.0%	93.6%	90.8%	0.87
Linear SVM	92.7%	99.3%	96.8%	0.95
Q-SVM	95.7%	100%	98.4%	0.97
C-SVM	94.7%	100%	98.0%	0.97

TABLE II. MAXIMUM ENTROPY-BASED FEATURES SELECTION RESULTS ON BRATS 2014

Classifiers	SE	SP	ACC	DSC
DT	92.5%	94.7%	94.0%	0.91
EBT	94.6%	95.1%	95.0%	0.92
LDA	94.8%	91.7%	92.6%	0.88
QDA	96.2%	89.7%	91.6%	0.86
Linear SVM	98.8%	96.1%	97.1%	0.95
Q-SVM	99.6%	99.8%	99.8%	0.99
C-SVM	99.6%	99.8%	99.8%	0.99

TABLE III. MAXIMUM ENTROPY-BASED FEATURES SELECTION RESULTS ON BRATS 2015

Classifiers	SE	SP	ACC	DSC
DT	98.8%	99.6%	99.4%	0.99
EBT	99.6%	99.4%	99.5%	0.99
LDA	100%	97.5%	98.2%	0.97
QDA	100%	97.5%	98.3%	0.97
Linear SVM	100%	78.9%	81.8%	0.60
Q-SVM	100%	98.0%	98.7%	0.97
C-SVM	92.9%	100%	97.6%	0.96

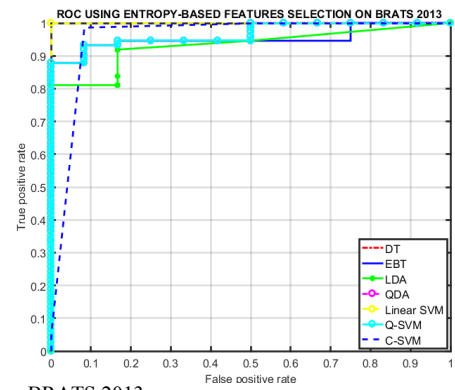


Fig. 4. ROC on BRATS 2013

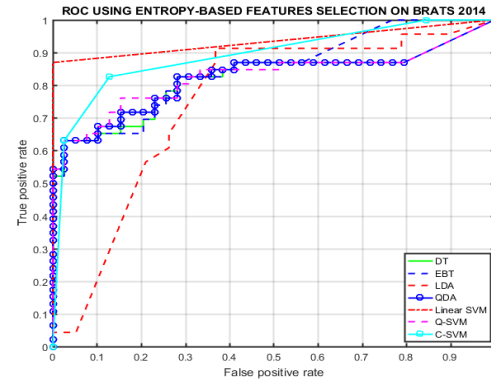


Fig. 5. ROC on BRATS 2014

TABLE IV. PROPOSED METHOD RESULTS ON DIFFERENT CROSS-VALIDATION APPROACH

5 fold					10 fold				
BRATS 2013					BRATS 2013				
Classifiers	SE	SP	ACC	DSC	Classifiers	SE	SP	ACC	DSC
DT	0.7547	0.8642	0.8160	0.7830	DT	84.6%	91.8%	89.2%	0.85
EBT	0.8264	0.8310	0.8292	0.7952	EBT	90.8%	99.3%	96.0%	0.94
LDA	0.7692	0.7813	0.7764	0.7353	LDA	89.7%	98.6%	95.2%	0.93
QDA	0.7692	0.7712	0.7704	0.7286	QDA	86.0%	93.6%	90.8%	0.87
Linear SVM	0.7117	0.8043	0.7645	0.7220	Linear SVM	92.7%	99.3%	96.8%	0.95
Q-SVM	0.7168	0.8108	0.7704	0.7286	Q-SVM	95.7%	100%	98.4%	0.97
C-SVM	0.7353	0.8219	0.7849	0.7449	C-SVM	94.7%	100%	98.0%	0.97
BRATS 2014					BRATS 2014				
DT	0.7424	0.5682	0.6667	0.7158	DT	92.5%	94.7%	94.0%	0.91
EBT	0.7456	0.5650	0.6667	0.7158	EBT	94.6%	95.1%	95.0%	0.92
LDA	0.7424	0.5882	0.6767	0.7249	LDA	94.8%	91.7%	92.6%	0.88
QDA	0.9524	0.6667	0.7843	0.7843	QDA	96.2%	89.7%	91.6%	0.86
Linear SVM	0.7979	0.8163	0.8060	0.8219	Linear SVM	98.8%	96.1%	97.1%	0.95
Q-SVM	0.7862	0.8667	0.8252	0.8224	Q-SVM	99.6%	99.8%	99.8%	0.99
C-SVM	0.7233	0.8235	0.7751	0.7566	C-SVM	99.6%	99.8%	99.8%	0.99
BRATS 2015					BRATS 2015				
DT	0.9259	0.8639	0.8837	0.8351	DT	98.8%	99.6%	99.4%	0.99
EBT	0.9217	0.8791	0.8929	0.8475	EBT	99.6%	99.4%	99.5%	0.99
LDA	0.9174	0.8602	0.8785	0.8282	LDA	100%	97.5%	98.2%	0.97
QDA	0.9048	0.8652	0.8779	0.8261	QDA	100%	97.5%	98.3%	0.97
Linear SVM	0.8491	0.8353	0.8410	0.8140	Linear SVM	100%	78.9%	81.8%	0.60
Q-SVM	0.8734	0.8621	0.8658	0.8114	Q-SVM	100%	98.0%	98.7%	0.97
C-SVM	0.7710	0.8642	0.8320	0.7604	C-SVM	92.9%	100%	97.6%	0.96

TABLE V. COMPARISON BETWEEN PROPOSED AND PREVIOUS TECHNIQUES

Ref	Year	Feature extraction method	Classifiers	MRI modality	Results
[41]	2017	Tamura and LBP	SVM	T2	100% SE, 97.5% SP and 98.90% Accuracy
[42]	2017	Hu moment invariants (HMI)	Twin SVM	T2	97.33% Accuracy
[43]	2017	Multi-fractal features	SVM	T2	98.01% $\pm$ 0.07 Accuracy
[44]	2017	Pseudo zernike moment	Linear regression	T2	97.51% Accuracy
[45]	2018	GLCM	ANFIS	T1, FLAIR, T2	92.03% Accuracy and 93.79% DSC
[46]	2017	Texture features	RF classifier	T1, FLAIR, T2, and T1c	59% Average DSC
Proposed method		Texture and shape based features	Multiple classifiers	T1, FLAIR, T2, and T1c	99.6% SE, 99.8% SP, 99.8% ACC, and 0.99 DSC

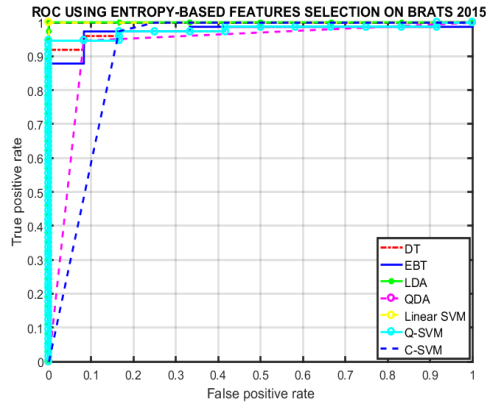


Fig. 6. Roc on BRATS 2015

Table IV mentions the comparison between the existing and proposed method. Figure 4,5,6 shows ROC on BRATS datasets. On the basis of performance evaluation, we observe that EBT obtains 99.5% accuracy on BRATS 2015 and Q-SVM, C-SVM achieves 99.8% accuracy on BRATS 2014. Q-SVM obtains 98.4% accuracy on BRATS 2013 dataset.

The proposed methods results are compared with the 5/10 fold cross-validation approach are reported in Table IV. Maximum 99.8% ACC achieves using 10 fold and 84.7%ACC achieve on 5 fold approach. The proposed method comparison with existing using 10-fold is reported in Table VI.

TABLE VI. PROPOSED METHOD COMPARISON USING THE 10-FOLD CROSS-VALIDATION APPROACH

REF	YEAR	CLASSIFICATION METHOD	RESULTS
[41]	2017	10 fold	98.90% ACC
[47]	2018	10 fold	98% ACC
[48]	2018	10 fold	99.69%ACC
PROPOSED		10 fold	99.8% ACC

III. CONCLUSION

In this work, a new feature fusion method is presented for brain tumor classification. The proposed method is comprised of the feature extraction, the entropy-based feature selection, and fusion approach is used for prediction between the benign and malignant images. The proposed method is separately trained with benchmark databases. The testing is performed based on training. The testing results depict 0.99 dice similarity coefficient (DSC), 99.6% sensitivity (SE), 99.8 % SP and 99.8% accuracy. These satisfactory results prove the robustness of the proposed approach.

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